NIPPON STEEL

NIPPON STEEL CORPORATION
KANSAI WORKS AMAGASAKI AREA
1 HIGASHIMUKOJIMA NISHINO-CHO, AMAGASAKI, JAPAN

INSPECTION CERTIFICATE

CERTIFICATE NO. :0YYE0915 PAGE : 1/2 DATE :2022-07-08

CUSTOMER :ENERGY METALS, INC.
ORDER NO. :P.O.306401-00
SHIPPER :SANKYO & CO., LTD. 111 5246-AMZ1753B 2-848-N3-5-7-A862-04
COMMODITY :SEAMLESS COLD FINISHED STAINLESS STEEL PIPE
STANDARD :ASTM A790-20 / ASME 2019 EDITION SA-790 UNS S31803
ASTM A790-20 / ASME 2019 EDITION SA-790 UNS S32205
SPECIFICATION :
MILL WORK NO. :0YYE0915 O.D.:NPS1 W.T.:SCH160 LENGTH:MIN.20feet MAX.24feet QUANTITY:90pcs.
TOTAL LENGTH:2063.76feet MASS:2662kg

HEAT NO. :F12B027 F222021
PRODUCTS PCS. :8 82

HEAT TREATMENT :SOLUTION TREATED(1922°F X 2min. W.Q.)

CHEMICAL COMPOSITION(%)

		*1	C *4	Si	Mn	P *4	S *4	Cr *3	Ni *3	Mo *3	N	*PI *2
SPEC. MIN.		L	-	-	-	-	-	220	45	30	14	35
MAX.		L	30	100	200	30	20	230	65	35	20	-
HEAT NO.			✓	✓	✓	✓	✓	✓	✓	✓	✓	
F12B027		L	24	35	126	18	0	226	55	34	18	37
F222021		L	22	35	131	18	0	226	55	33	18	36

*1 L:LADLE ANALYSIS *2:X1 *3:X10 *4:X1000 OTHER:X100 *PI PITTING INDEX:Cr+3.3Mo+16N

TENSILE TEST

		*1	*2	YS *3		TS *3		EL %
SPEC. MIN.		L	B	P	65.0	P	95.0	25
MAX.		L	B	P	-	P	-	-
HEAT NO.					✓		✓	✓
F12B027		L	B	P	92.1	P	120.7	44
F222021		L	B	P	90.8	P	119.1	46

TYPE OF SPECIMEN:FULL SECTION *1 SAMPLING DIRECTION L:LONGITUDINAL *2 SAMPLING POSITION B:BASE METAL
*3 UNIT P:ksi GAUGE LENGTH:2.0in. KIND OF YS:0.2% OFFSET

HARDNESS TEST(HRC)

		*1	*2	*3
SPEC. MIN.		B	C	-
MAX.		B	C	28
HEAT NO.				(1) ✓
F12B027		B	C	25
		B	C	24
F222021		B	C	24
		B	C	24

*1 SAMPLING POSITION B:BASE METAL *2 LOCATION C:CROSS SECTION *3 EACH (1): IMPRESSION NO.

WE HEREBY CERTIFY THAT THE MATERIAL DESCRIBED HEREIN HAS BEEN MADE IN ACCORDANCE WITH THE RULES OF THE CONTRACT.

ACCEPTED
BY JR DATE 1/15/26

NOBUYUKI HISAMUNE
SENIOR MANAGER
QUALITY ASSURANCE DEPARTMENT

NO. 353870A22

<Notice>
Modification to or unauthorized use of this certificate is strictly prohibited. If you have any questions on the authenticity of this certificate, you can contact us by E-mail, pipe-ipp@jp.nipponsteel.com .

Job 089-25 PO: 40301 Lines: 011 + 012 HT: F22021

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FERRITE CONTENT TEST (ASTM E562) (%)

SPEC. MIN.			35.00
MAX.			55.00
HEAT NO.			
F12B027			40.10
F222021			43.70

CORROSION TEST(ASTM A923-C, 77°F X 24hr)
CORROSION RATE(MAX. 10mdd): 1mdd NO PITTING (X20) (F12B027)
: 1mdd NO PITTING (X20) (F222021)
CORROSION TEST(ASTM G48-A, 77°F X 24hr)
CORROSION RATE(MAX. 4.0g/m²): 0.2g/m² NO PITTING (X20) (F12B027)
: 0.2g/m² NO PITTING (X20) (F222021)
MICRO STRUCTURE TEST(ASTM A923-A): ACCEPTABLE
FLATTENING TEST: ACCEPTABLE
VISUAL & DIMENSIONS: ACCEPTABLE
HYDROSTATIC TEST 2500psi: ACCEPTABLE
POSITIVE MATERIAL IDENTIFICATION TEST(SPARK TEST): ACCEPTABLE
NACE MR0175/ISO 15156, NACE MR0103/ISO 17945 HARDNESS: GUARANTEED
CERTIFIED ACCORDING TO PED2014/68/EU, ANNEX I, PAR. 4.3 BY TUEV RHEINLAND INDUSTRIE SERVICE GmbH
(NOTIFIED BODY, ID-No. 0035/CERTIFICATE No. 01 202 J/Q-02 0008)
MERCURY WOULD NOT BE DETECTED
COUNTRY OF ORIGIN: JAPAN
EN 10204 3.1

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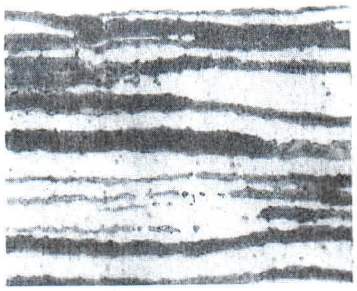
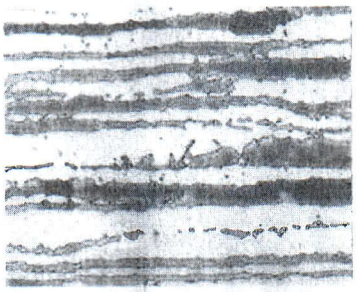
FERRITE CONTENT TEST (ASTM E562)

DOCUMENT NO. : 0YYE0915C
CERTIFICATE NO. : 0YYE0915

DATE: 2022-07-08

HEAT NO.	X 400
F12B027	
F222021	

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